METRIC
FF-B-171/32
30 NOVEMBER 1993

# FEDERAL SPECIFICATION SHEET

## BEARING, BALL, ANNULAR, DOUBLE ROW, ANGULAR CONTACT, NON-FILLING SLOT, VERTEX OF CONTACT ANGLES OUTSIDE BEARING, ANSI/AFBMA DIMENSION SERIES 32.

The complete requirements for acquiring the bearings described herein shall consist of this document and the latest issue of FF-B-171/GEN.

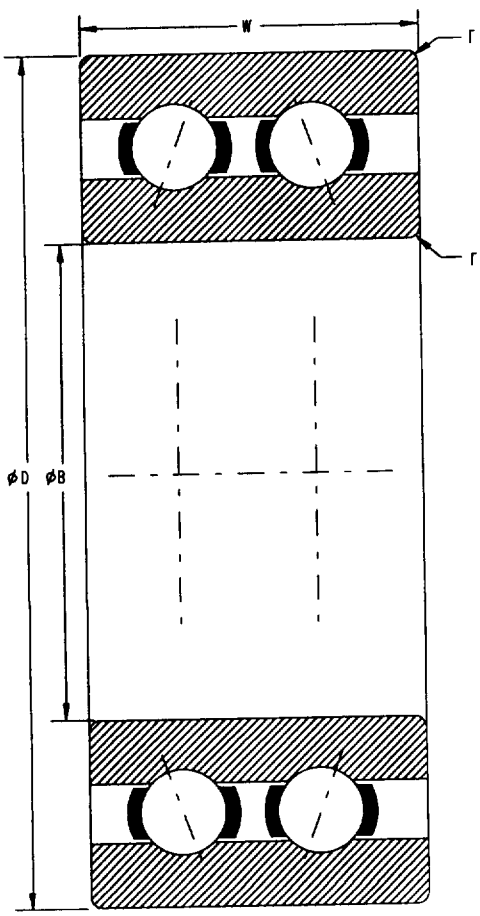
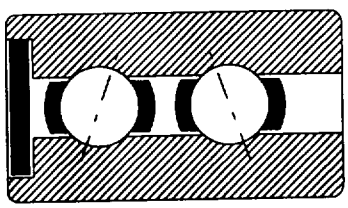
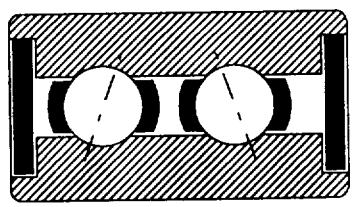
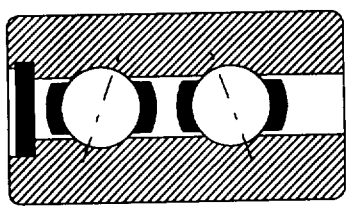
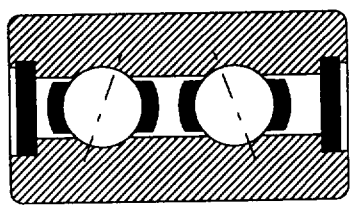


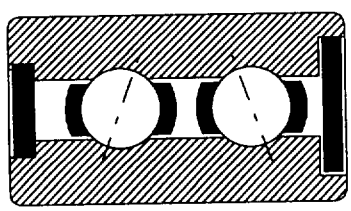
FIGURE 1.
OPEN (NO SHIELDS OR SEALS)
(CODE A)


FIGURE 2.
SINGLE SHIELD
(CODE B)


FIGURE 3.
DOUBLE SHIELD
(CODE C)


FIGURE 4.
SINGLE SEAL
(CODE D)


FIGURE 5.
DOUBLE SEAL
(CODE E)


FIGURE 6.
SINGLE SHIELD AND SEAL
(CODE F)

BEARING CONFIGURATIONS

AMSC/NA

FSC 3110

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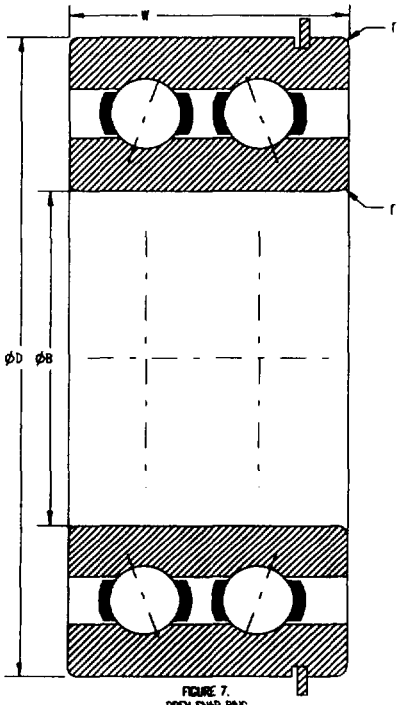



FIGURE 7.
OPEN SNAP RING
(CODE G)

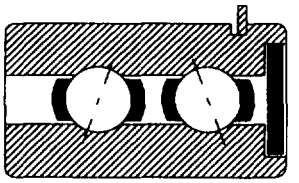


FIGURE 8.
SINGLE SHIELD-
ATTACHED ON SNAP
RING SIDE
(CODE H)

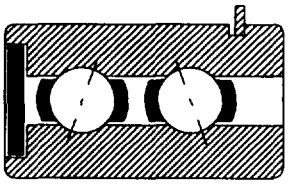


FIGURE 9.
SINGLE SHIELD-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE J)

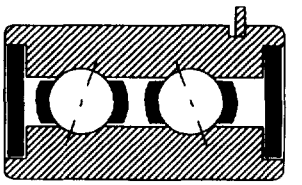


FIGURE 10.
DOUBLE SHIELD-WITH
SNAP RING
(CODE K)

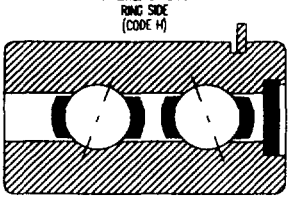


FIGURE 11.
SINGLE SEAL-
ATTACHED ON SNAP
RING SIDE
(CODE L)

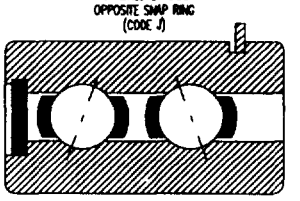


FIGURE 12.
SINGLE SEAL-
ATTACHED ON SIDE
OPPOSITE SNAP RING
(CODE M)

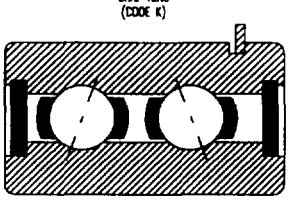


FIGURE 13.
DOUBLE SEAL-WITH
SNAP RING
(CODE N)

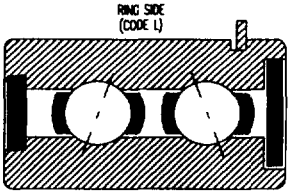


FIGURE 14.
SINGLE SHIELD AND SEAL-
SHIELD ON SNAP RING SIDE
(CODE P)

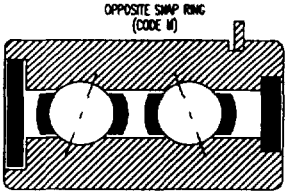


FIGURE 15.
SINGLE SHIELD
AND SEAL-SEAL
ON SNAP RING SIDE
(CODE R)

BEARING CONFIGURATIONS - (Continued)



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REQUIREMENTS:

1. Boundary dimensions. The boundary dimensions shall be specified in the part number by the size codes in Table I.

2. Cages. Cages shall be manufactured from any material unless otherwise specified. Cages shall be specified in the part number by the codes in table II.

3. Shields, seals, and snap rings. Shields and seals shall be fabricated from any material unless otherwise specified and shall be removable or nonremovable type. Seals shall be impervious contact type. Snap rings shall be fabricated from ferrous material unless otherwise specified and shall have sufficient spring temper to set firmly in the ring grooves and shall be removable without the metal taking a permanent set. The snap ring and snap ring groove dimensions and their relative locations shall be in accordance with ANSI/AFBMA Standard 20. The number and positioning of shields, seals, and snap rings required shall be specified in the part number by the codes in table III.

4. Radial Internal Clearance. The tolerance limits for radial internal clearance shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

5. Tolerance Class. The tolerance limits for tolerance class ABEC-1 and ABEC-5 shall be in accordance with ANSI/AFBMA Standard 20 and shall be specified in the part number by the codes in table IV.

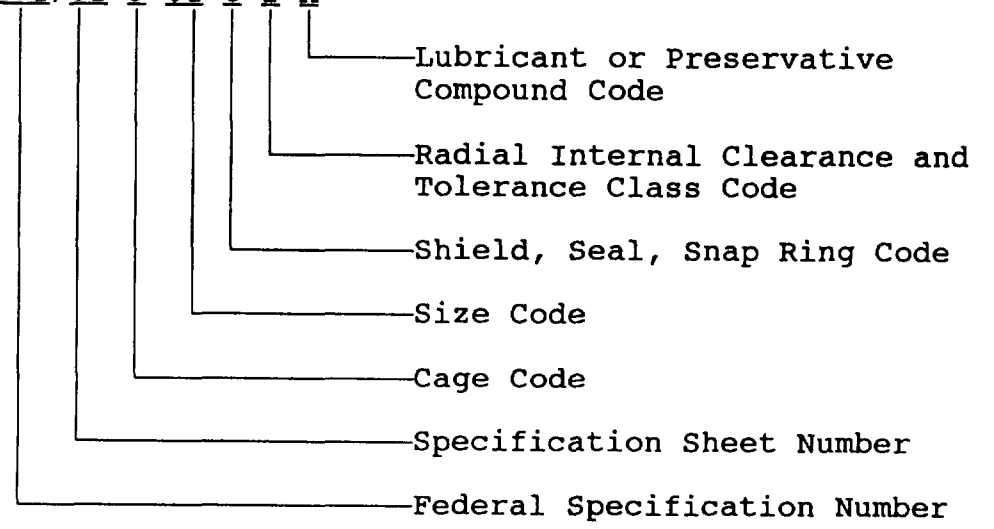
6. Lubricant or Preservative Compound. Preservative compound shall be applied to all open bearings (no shields or seals) and to single closure bearings (single shield or single seal) that have no additional lubricant requirement. Thus, for these bearings the bearing part number shall include the "C" suffix. For double closure or single closure bearings that require lubricant the bearing part number shall include the "A" or "B" suffix. When grease is required, the bearing void (the airspace between the inner and outer ring of the assembled bearing) shall be 25 to 50 percent filled. The lubricant or preservative compound shall be specified in the part number by the codes in table V.



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7. Part number. The part number consists of the letters FFB, the basic Federal specification number, the specification sheet number, and alpha or numeric codes for cage, size, shields, seals, and snap rings, radial internal clearance and tolerance class, and lubricant or preservative compound.

EXAMPLE: FFB171/32 J 01 C 2 A




FFB171/32J01C2A indicates - Bearing, cage-pressed steel or molded plastic; bore 12mm; outside diameter 32mm; width 15.9mm; two shields; radial internal clearance symbol 0 (normal); tolerance class ABEC-1; grease in accordance with MIL-G-81322.



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TABLE I. Boundary dimensions.

| Size Code 1/ | OB Bore | OD Outside Diameter | W Width | Chamfer $r_s$min 2/ | Shaft Shoulder Diameter Min 3/ | C Dynamic load rating (pounds) |
|---|---|---|---|---|---|---|
| 00 | 10 | 30 | 14.3 | 0.6 | 15 | 1320 |
| 01 | 12 | 32 | 15.9 | 0.6 | 17 | 1730 |
| 02 | 15 | 35 | 15.9 | 0.6 | 20 | 1970 |
| 03 | 17 | 40 | 17.5 | 0.6 | 22 | 2170 |
| 04 | 20 | 47 | 20.6 | 1.0 | 26 | 3140 |
| 05 | 25 | 52 | 20.6 | 1.0 | 31 | 3400 |
| 06 | 30 | 62 | 23.8 | 1.0 | 36 | 4720 |
| 07 | 35 | 72 | 27.0 | 1.1 | 42 | 6230 |
| 08 | 40 | 80 | 30.2 | 1.1 | 47 | 7480 |
| 09 | 45 | 85 | 30.2 | 1.1 | 52 | 7985 |
| 10 | 50 | 90 | 30.2 | 1.1 | 57 | 7925 |
| 11 | 55 | 100 | 33.3 | 1.5 | 63.5 | 9350 |
| 12 | 60 | 110 | 36.5 | 1.5 | 68.5 | 11445 |
| 13 | 65 | 120 | 38.1 | 1.5 | 73.5 | 13325 |
| 14 | 70 | 125 | 39.7 | 1.5 | 78.5 | 14540 |
| 15 | 75 | 130 | 41.3 | 1.5 | 84 | 15805 |
| 16 | 80 | 140 | 44.4 | 2.0 | 90 | 17380 |
| 17 | 85 | 150 | 49.2 | 2.0 | 95 | 20105 |
| 18 | 90 | 160 | 52.4 | 2.0 | 100 | 21460 |
| 19 | 95 | 170 | 55.6 | 2.1 | 107 | 26080 |
| 20 | 100 | 180 | 60.3 | 2.1 | 112 | 29320 |

1/ Size codes for Table I are industry bore code numbers.
2/ The minimum bearing chamfer ($r_s$min) must clear the maximum shaft or housing fillet radius. This dimension does not control the shape of the bearing chamfer contour.
3/ For reference purposes only. Maximum shoulder heights shall be such as to permit bearing removal pressure to be applied to the bearing inner ring. In cases where this requirement cannot be met, provisions shall be made for the removal of the bearing from the shaft without damage to the bearing or other parts.

TABLE II. Codes for Cages.

| Code | Cage |
|---|---|
| J | Pressed steel or molded plastic |
| M | Machined bronze or brass |
| T | Machined non-metallic (phenolic) |
| Y | Pressed brass |



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TABLE III. Codes for Shields and Seals.

| Code | Shield, seal, and snap ring configuration |
|---|---|
| A | open (no shields or seals) - fig.1 |
| B | single shield - fig.2 |
| C | double shield - fig.3 |
| D | single seal - fig.4 |
| E | double seal - fig.5 |
| F | single shield and seal - fig.6 |
| G | open with snap ring - fig.7 |
| H | single shield (attached on snap ring side) - fig.8 |
| J | single shield (attached on side opposite snap ring)-fig.9 |
| K | double shield with snap ring - fig.10 |
| L | single seal (attached on snap ring side) - fig.11 |
| M | single seal (attached on side opposite snap ring)-fig.12 |
| N | double seal with snap ring - fig.13 |
| P | single shield and seal (shield on snap ring side)-fig.14 |
| R | single shield and seal (seal on snap ring side) - fig.15 |

TABLE IV. Codes for Radial Internal Clearance and ABEC tolerance.

| Code | Radial Internal Clearance | Tolerance Class |
|---|---|---|
| 1 | Symbol 2 (less than normal) | ABEC-1 |
| 2 | Symbol 0 (normal) | |
| 3 | Symbol 3 (greater than normal) | |
| 4 | Symbol 4 (greater than symbol 3) | |
| 5 | Symbol 2 (less than normal) | ABEC-5 |
| 6 | Symbol 0 (normal) | |
| 7 | Symbol 3 (greater than normal) | |
| 8 | Symbol 4 (greater than symbol 3) | |

TABLE V. Codes for Lubricant and Preservative Compound.

| Code | Lubricant or Preservative Compound |
|---|---|
| A | Grease in accordance with MIL-G-81322 |
| B | Grease in accordance with DOD-G-24508 |
| C | Preservative compound in accordance with MIL-P-197 |



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MILITARY INTERESTS:

Military Coordinating Activity:
DLA - IS

Custodians:
Army - AT
Air Force - 11

Review Activities:
Army - AR, EA, MI
Navy - OS

User Activity:
Army - GL, ME

CIVIL AGENCY COORDINATING ACTIVITIES:
GSA-FSS

PREPARING ACTIVITY:
DLA-IS

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